cadence

HPNFC Firmware Driver Quick Start Guide

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1. Overview

This document describes the basics of how to get up and running with the Cadence IP Core Driver.

- Prepare the Cadence Platform Services (CPS)
- Prepare interrupt handling
- Build the core-driver
- Use the core-driver to initialize the Cadence IP

2. Details

Here's how to get started:

1. *Prepare the Cadence Platform Services:*

   This is covered in detail in the Cadence Driver Porting Guide, which is available at doc/porting/porting_guide.pdf. A sample implementation for bare-metal systems is available in RefCode/init/cps_bm.c . You may be able to use these without modification.

   CPS is a platform-specific code to connect a given Cadence driver to the rest of your system. Not all of the features of CPS are required for this driver. You only need to implement the functions for uncached access to memory:

   ```
   CPS_UncachedRead8, CPS_UncachedRead16, CPS_UncachedRead32,
   CPS_UncachedWrite8, CPS_UncachedWrite16 and CPS_UncachedWrite32
   ```

   These functions will be used by the core driver to access the memory in a portable manner. CPS_CacheFlush() / CPS_CacheInvalidate() are not used by this driver. However, this means that you must ensure cache consistency of any areas of memory passed to the driver for use in DMA transfers.

   After this step you should have an object file implementing the CPS for your target system which you can link with later.

2. *Build the core-driver*

   The driver and a sample makefile is provided in the CoreDriver directory. You will need to customize the makefile for your build environment, e.g. compiler, any necessary libraries etc.

   After this step you should have the library file libhpnfc.a which can be linked with your own code.

3. *Use the core-driver to initialize the Cadence IP*

   Sample code is available in the RefCode directory.

   The sample code is designed to require only minor changes before it can be used. You need to implement main function and call InitDriver function with proper registers addresses. You probably need to implement your own slaveDmaHandler function to handle Slave DMA data transfers. After that you can execute other operations on the driver.

   The sample code then calls *probe* to confirm that the hardware is available at the specified address, and to determine the size of memory which is required for private data. This memory must be provided by the code calling the driver, and may be allocated at compile time (e.g. by an array allocation as shown in the sample code) or dynamically. This private data will be passed to the driver as a parameter for each call to the API.

   The sample code then calls *init* to initialize the driver. To the init function

   As well as initializing the driver, the sample code also displays some diagnostic information using stdio. If stdio is not available on your system, these calls should be stubbed out or replaced with an alternative (e.g. write to memory or local storage.)

   After modifying the sample code to suit your environment, you should compile for your target system and link with the driver and CPS objects prepared previously, then download and run the code on your target system. This will be hardware and operating system specific, and is not covered in this documentation.